The Shrinking Caspian Sea: Eco-hydrological Destabilization under Compound Human and Climate Stressors

Jesse Duku[1], Amir AghaKouchak[1,2]

[1]Department of Civil & Environmental Engineering, Center for Hydrometeorology & Remote Sensing, University of California, Irvine, CA, USA

[2]United Nations University Institute for Water, Environment and Health (UNU-INWEH), Richmond Hill, ON, Canada

Abstract

The Caspian Sea has undergone a sustained multidecadal water-level decline, but the relative roles of hydrological and climatic drivers remain uncertain. In this study, we examine satellite observations, reanalysis datasets, and long-term river discharge records to assess key components of the Caspian hydrological system. Analyses show a notable decrease in total river inflow dominated by the Volga River, while precipitation over the lake and basin shows weak or insignificant long-term trends. Evaporation estimates from ERA5 indicate a modest increasing tendency consistent with regional warming, though uncertainties remain over lake-surface fluxes. These findings reveal a shift in the water balance of the basin driven by compound human and climatic stresses. Preliminary ecological indicators, including chlorophyll-a trends, suggest changing conditions in the northern Caspian. Without urgent intervention, the Caspian Sea may become a sobering example of irreversible anthropogenic desiccation.

Background

- The Caspian Sea is the world's largest inland water body with no surface outflow.
- Water level has declined persistently over the past 30 years.
- Drivers remain contested: Climate Variability VS Human water use VS Ecological change.
- This study evaluates key hydrologic components using long-term observations.

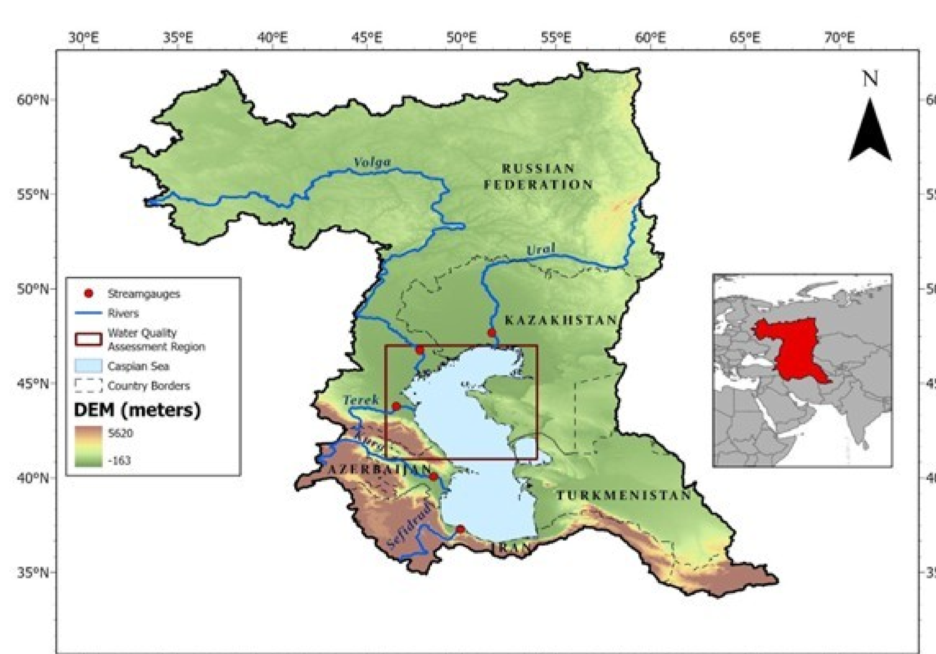

Fig.1 Map of the Caspian Hydrological Basin

Data and Methods

- **Hydrological datasets:**
  - **P:** GPCC precipitation (1991–2023).
  - **E:** ERA5 evaporation.
  - River discharge: Volga, Ural, Terek, Sefidrud, Kura.
  - Lake level: satellite altimetry.
  - TWS: GRACE / GRACE-FO.
- **Ecological variable:**
  - MODIS chlorophyll-a (northern Caspian)
- **Analysis:**
  - Long-term trend analysis
  - Comparison with observed Caspian Sea level decline

Note: Mass-balance attribution is part of ongoing work

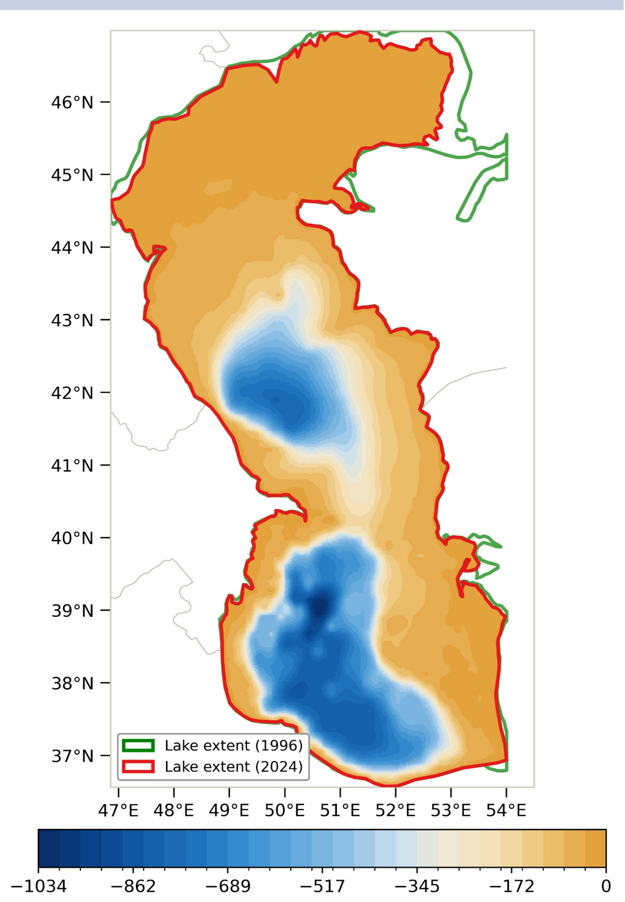

Fig 2. Spatial distribution of the Caspian Sea's bathymetry with lake area changes in 1996(green) and 2024(red) overlaid. Estimated loss of 23,858 km2 (5.47 %) of surface area.

Key Results

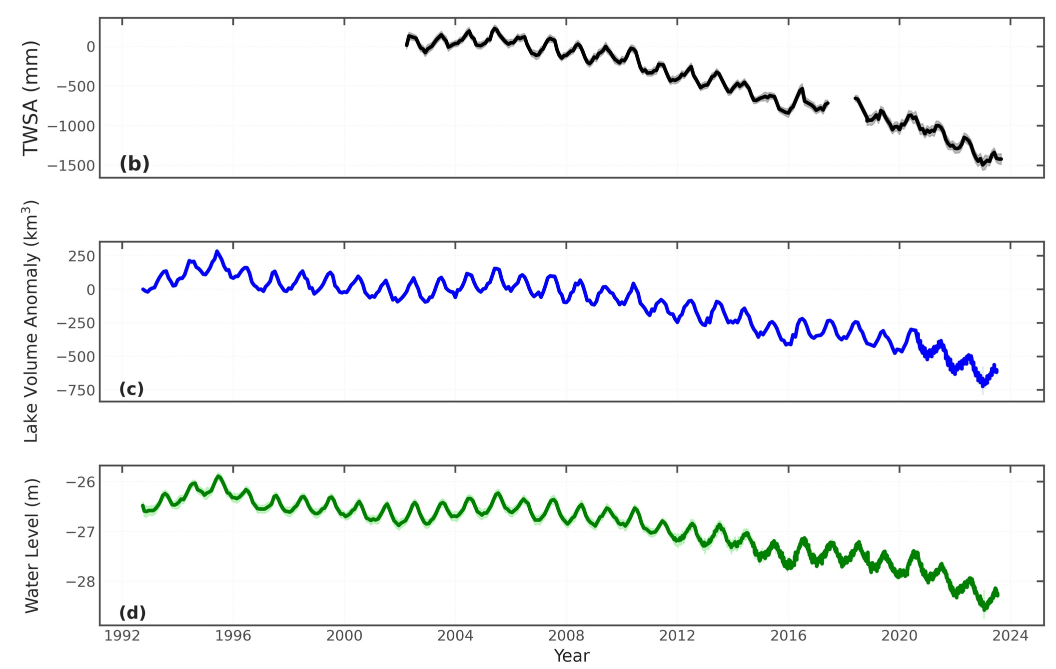

Fig.3. Total Water Storage (TWS) Anomaly (mm water equivalent) from GRACE/GRACE-FO showing a negative trend. (c) Lake volume anomaly (km3). (d) Caspian Sea water height (m) below mean sea level.

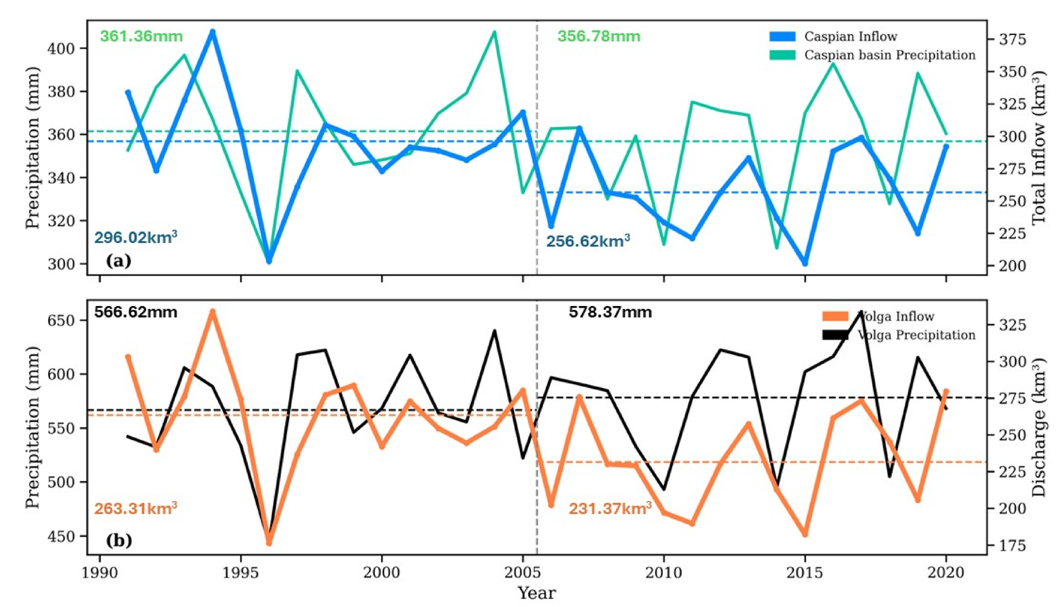

Fig.4. Hydrological Inputs to the Caspian Sea (a) Caspian Sea total inflow (blue line, km³/year) and precipitation (green line, mm/year). (b) Volga precipitation (black line, mm/year) and discharge (orange line, km³/year).

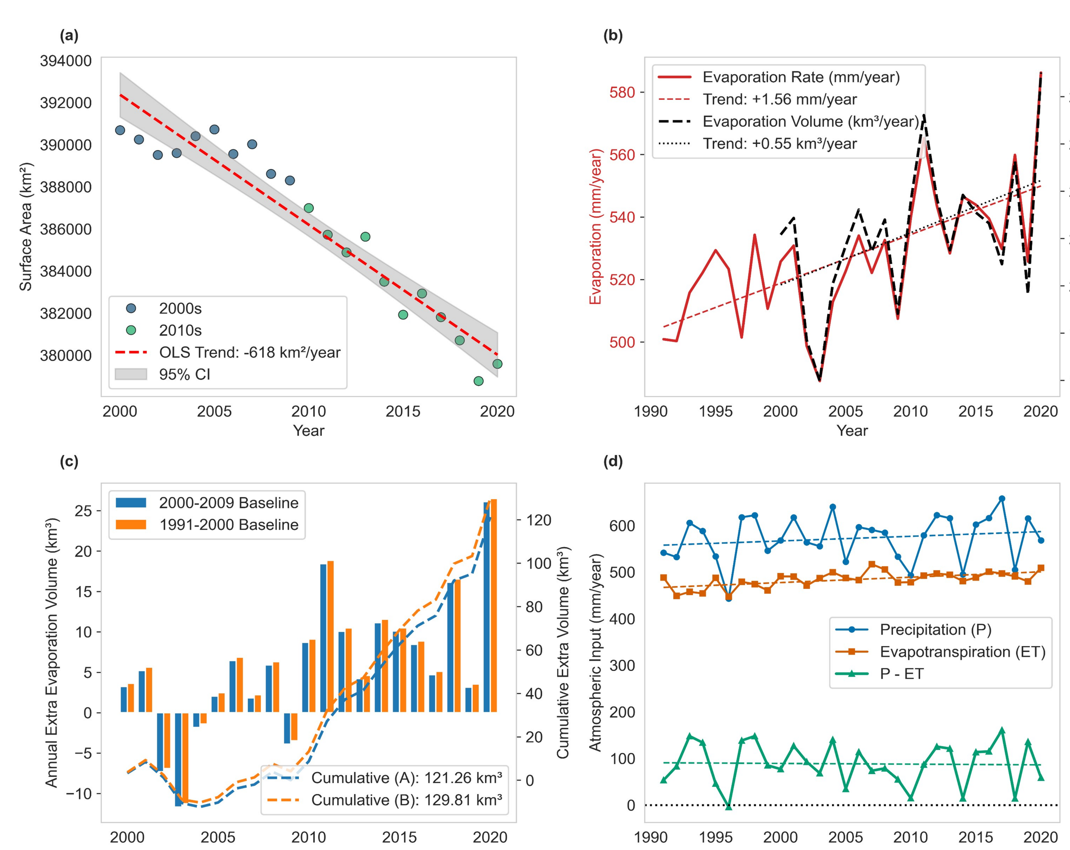

Fig.5. Evaporation flux and its contribution to the Caspian Sea water loss (1991–2020). (a) Annual median surface area of the Caspian Sea showing a decline of ~ 618km2/yr. (b) Evaporation rate (red, mm/yr) and evaporation volume (black, km3/yr) with linear trends over the Caspian Sea showing evaporation increases over the sea. (c) Annual and Cumulative ET relative to two baselines (A: 2000-2009; B: 1991-2000). Cumulative excess evaporation explains 19~20% of observed volume loss. (d) Annual precipitation (blue), Evaporation (orange), and net atmospheric Input over the Volga basin (green).

Key Results

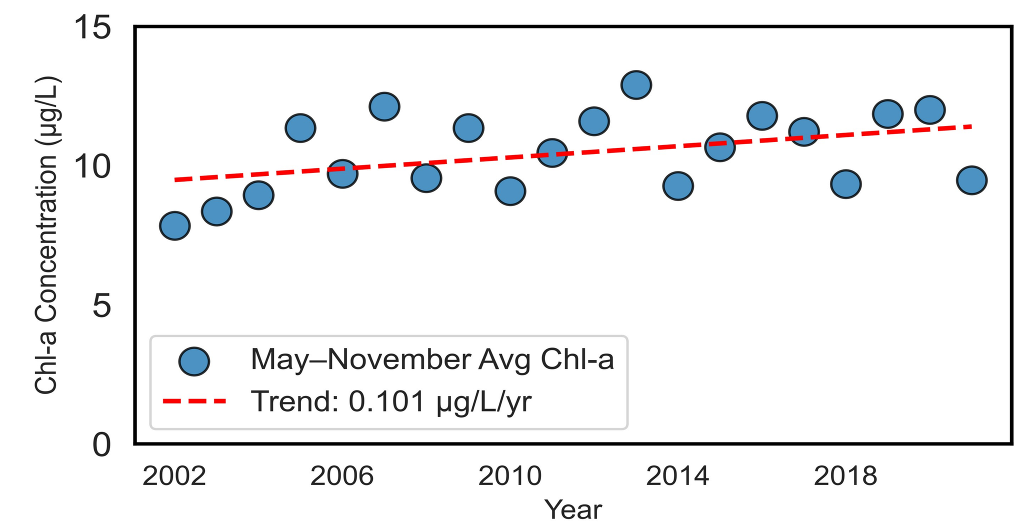

Fig.6. Annual average chlorophyll-a (chl-a) concentrations in the northern Caspian Sea from 2002 to 2021 from MODIS L3SMI. Over these 20 years, the average Chl-a concentration ranged from 8 to 13 µg/L, with an upward trend of approximately 1 µg/L per decade

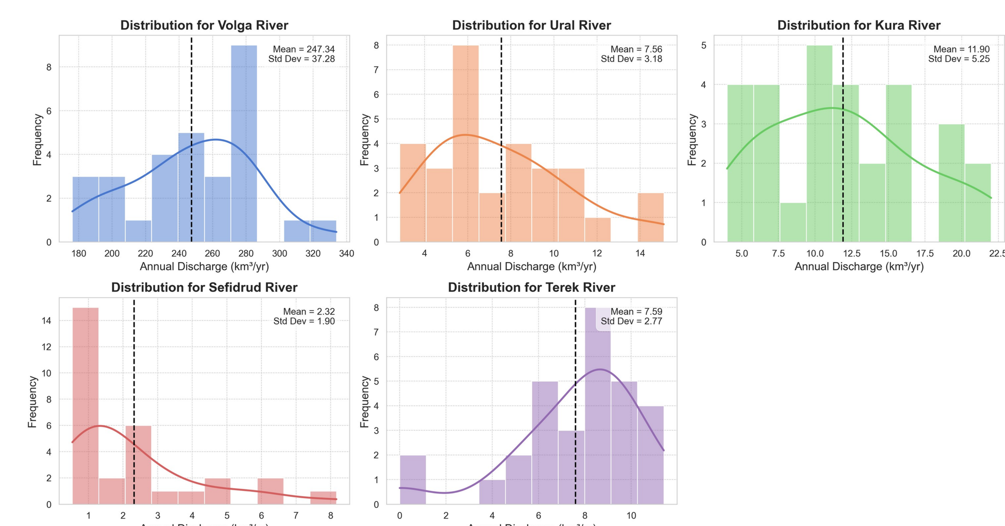

Fig.5. Histograms for the differences between river discharge from the five main littoral countries surrounding the Caspian Sea. The Volga alone contributes 90 % of the river inflow to the Caspian Sea.

Conclusion & Future Work

- Long-term observations confirm a sustained multidecadal decline in Caspian Sea water level, coinciding with reductions in total river inflow, especially from the Volga River, and modest increases in evaporation.
- Precipitation over both the lake and the broader basin shows weak or insignificant long-term trends, indicating limited contribution from atmospheric input changes to the observed decline.
- Development of a **full water-balance reconstruction** ($\Delta H = P - E + R \pm \varepsilon$) to quantify the contributions of each hydrologic component over time.

References

- R. Samant, M. Prange, Climate-driven 21st century Caspian Sea level decline estimated from CMIP6 projections. Commun. Earth Environ. 4, 1–12 (2023).
- A. AghaKouchak, et al., Aral Sea syndrome desiccates Lake Urmia: Call for action. J. Gt. Lakes Res. 41, 307–311 (2015).

Acknowledgments

This project is funded by NSF Accelnet Award Number 2114701, managed by the National Academies of Sciences, Engineering, and Medicine.

For further information

Contact jduku@uci.edu for more information

AGU25 New Orleans, LA | 15–19 December 2025 WHERE SCIENCE CONNECTS US